

Assignment - 1

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SECTION : SB(1)

ROLL NO : 9
SUBJECT : FULL STACK USING SCRIPTING TECHNOLOGIES

Q1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

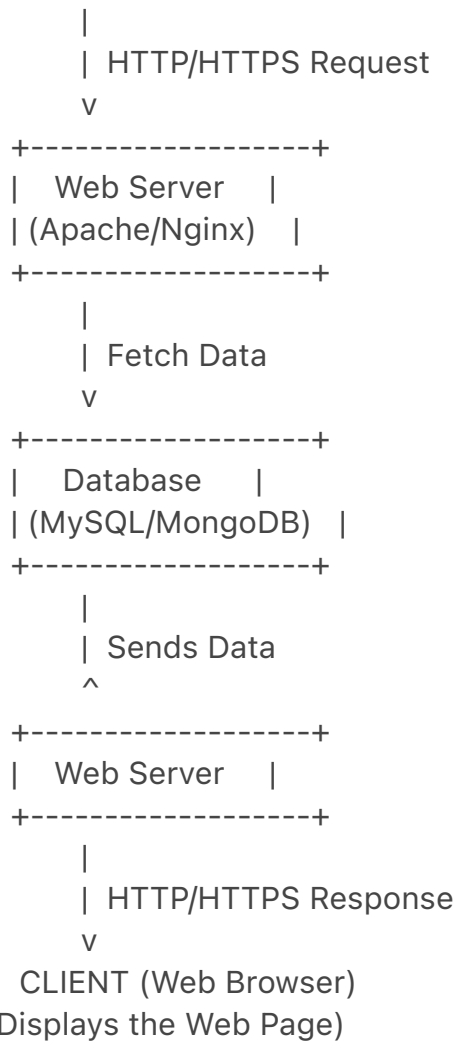
Ans :

Feature	Frontend Development	Backend Development	Full-Stack Development
Definition	Develops the part of a website that users see and interact with.	Develops the server-side logic, database, and APIs that power the website.	Works on both the frontend and backend of a website.
Main Technologies	HTML, CSS, JavaScript, React, Angular, Vue	Java, Python, PHP, Node.js, C#, MySQL, MongoDB	Combination of frontend and backend technologies
Focus	User Interface (UI) and User Experience (UX)	Business logic, security, authentication, and database management	Builds the complete web application from start to finish
Example Tasks	Designing web pages, buttons, forms, animations	Managing user accounts, storing data, processing payments	Creating both the website interface and the server-side functionality

Q 2. Create a simple diagram showing how the client-server model works in web architecture.

Ans : CLIENT (Web Browser)

(Chrome, Firefox, Edge)



Q 3: Describe how a browser requests and displays a web page from a web server.

Ans : When we open a website, the browser and web server work together to show the web page.

Steps:

1. The user types the website address (URL) in the browser.
2. The browser sends a request to the web server.
3. The web server receives the request.
4. The server finds the required web page or data.
5. The server sends the web page back to the browser.
6. The browser reads the files and displays the web page on the screen.

Q 4: Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Ans : To start web development, we need some basic tools. These tools help us write, test, and manage our code.

Tool	Purpose
Visual Studio Code (VS Code)	It is a code editor used to write HTML, CSS, and JavaScript code.
Web Browser (Google Chrome, Firefox)	It is used to open and test web pages.
Live Server	It shows the web page in the browser and updates it automatically after saving changes.
Git	It is used to keep track of changes in the project.
GitHub	It is used to store and share projects online.
Node.js	It allows JavaScript to run outside the browser and helps install useful packages.

Q 5: Explain what a web server is and give examples of commonly used servers.

Ans : A **web server** is a computer or software that stores website files and sends them to the user's browser when requested.

When a user opens a website, the browser sends a request to the web server. The web server processes the request and sends the required web page back to the browser.

Examples of Commonly Used Web Servers:

1. **Apache HTTP Server** – One of the most popular web servers used for hosting websites.
2. **Nginx** – A fast and powerful web server used for high-traffic websites.
3. **Microsoft IIS (Internet Information Services)** – A web server developed by Microsoft for Windows.

LiteSpeed – A high-performance web server used by many web hosting companies.

Q 6 : Define the roles of a frontend developer, backend developer, and database administrator in a project.

Ans : In a web development project, different people have different responsibilities.

1. Frontend Developer

A frontend developer creates the part of the website that users can **see and use**. They design the layout, buttons, menus, and web pages.

Example: Creating the homepage and login page of a website.

2. Backend Developer

A backend developer works **behind the scenes**. They create the server-side logic, connect the website to the database, and manage user data.

Example: Checking the username and password when a user logs in.

3. Database Administrator (DBA)

A Database Administrator (DBA) manages the **database**. They store, organize, and protect the data. They also make sure the database works properly.

Example: Managing customer details, product information, and order records in an online shopping website.

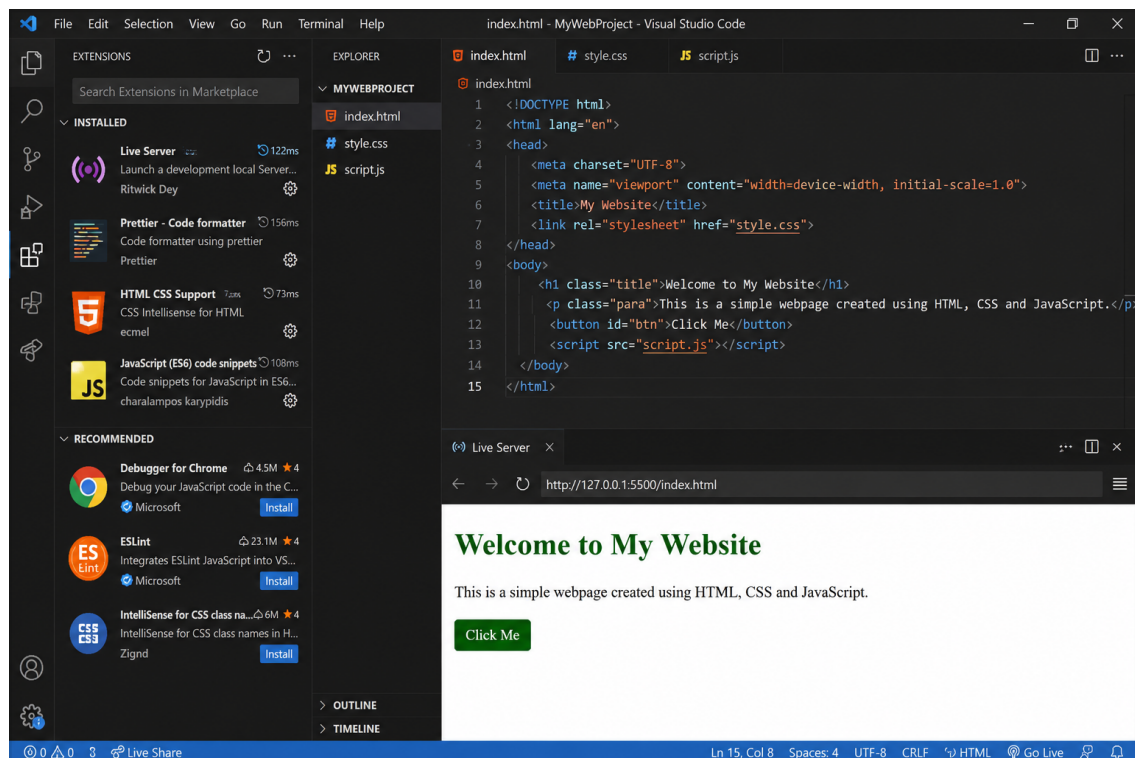
Conclusion

- **Frontend Developer** – Creates the part of the website that users can see and use.
- **Backend Developer** – Handles the server, application logic, and data processing.

Database Administrator (DBA) – Manages and secures the database.

Q 7 : Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Ans :



Q 8 : Explain the difference between static and dynamic websites. Provide an example of each.

Ans : **Difference Between Static and Dynamic Websites**

Static Website	Dynamic Website
Shows the same content to every user.	Shows different content to different users.
Easy to create and maintain.	More complex to develop.
Does not use a database.	Uses a database to store and manage data.
Loads faster.	May take slightly more time because it processes data.

Example: A personal portfolio website or a college information website. (Static Website).

Example: Amazon, Facebook, Instagram, and YouTube.(Dynamic Website)

Q 9 : Research and list five web browsers. Explain how rendering engines differ between them.

Ans : A **web browser** is a software application used to open and view websites on the internet.

Five Popular Web Browsers

1. **Google Chrome**
2. **Mozilla Firefox**
3. **Microsoft Edge**
4. **Apple Safari**
5. **Opera**

Rendering Engines Used by These Browsers

Web Browser	Rendering Engine
Google Chrome	Blink
Mozilla Firefox	Gecko
Microsoft Edge	Blink
Apple Safari	WebKit
Opera	Blink

How Rendering Engines Differ

A **rendering engine** is a software component that reads **HTML, CSS, and JavaScript** and displays the web page on the screen.

- **Blink** is fast and is used by **Google Chrome, Microsoft Edge, and Opera**.
- **Gecko** is used only by **Mozilla Firefox** and is known for good web standards support.
- **WebKit** is used by **Apple Safari** and is designed to work well on Apple devices like Mac, iPhone, and iPad.

Conclusion

Different web browsers use different rendering engines, but all of them have the same job: **to display web pages correctly and quickly for the user.**

Q 10 : Draw a labeled diagram showing the basic web architecture flow - client, server, database, and APIs.

Ans : Basic Web Architecture Flow

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+-----+
|   Client   |
| (Web Browser/User) |
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